Echo Growth Analysis

Growth Modeling and Data Driven Strategy Group 6 November 2026

Proposed decision

We recommend keeping normal notifications and not rolling out the aggressive version. The experiment shows more app opens, but less cumulative listening and lower later retention. It does not demonstrate an ad revenue gain. A new experiment can assess a capped notification frequency and user settings.

Echo is a fictional podcast app funded by advertising, with an approximate CPM of €9 per thousand impressions [1]. The files cover January 2025 to June 2026. The data pack describes random assignment of new signups from 15 January to 15 March 2026. We compare 1,366 control users with 1,343 aggressive notification users, retaining everyone assigned, including those who become inactive.

Experiment results

The primary follow-up covers twelve complete weeks after the signup week, W1-W12. Follow-up ends between 12 April and 7 June depending on signup date. The assignment period therefore differs from total follow-up. Groups have similar platform and channel distributions. Age differences remain visible in the notebook [3].

Follow-up measure	Control	Aggressive	Interpretation
Known listening per assigned user	680.20 min	644.51 min	-5.25%
Sessions per assigned user	29.07	31.60	+8.71%
Ad revenue per assigned user	€0.865	€0.876	Uncertain gain
W4 retention	78.70%	78.03%	Uncertain gap
W8 retention	62.23%	52.05%	-10.18 points
W12 retention	51.32%	39.61%	-11.71 points

The primary difference is -35.69 minutes per person, with a 95% confidence interval from -62.83 to -9.61 minutes. Estimating the twenty missing durations and temporarily excluding the twenty affected users support the same reading. The cleaned CSV files remain unchanged. Intervals use 5,000 user-level bootstrap samples. Secondary outcomes are not adjusted for multiple comparisons.

The claimed +30% is plausible: a reconstruction from 15 January to 15 March gives +31.5% total sessions. It assumes activity starts after signup for the first partial week. This does not establish the Growth team's exact calculation. More opens during recruitment are not enough to demonstrate lasting value.

Diagnosis and data quality

Weekly activity means at least one session. Activation is at least twenty minutes of listening during W0, the signup week. This is a convention for initial usage, not a condition for W4 retention. Ten minutes classifies almost everyone as activated. Adding two sessions contributes little. W0 may be partial and episode-level durations are unavailable.

Among 19,230 signups with complete W4 follow-up, 15,148 are active in W4, or 78.8%. Each person counts once. The funnel before signup is unobserved. Following a show, receiving a notification, listening and returning form a plausible loop, without direct referral measurement.

Channels and ad revenue value

Channel	Adjusted W4 retention	12-week revenue	52-week revenue	CAC
Paid social	71.4%	€0.843	€1.615	€2.608
Organic store	83.3%	€0.991	€2.554	Unknown
Podcast cross-promo	86.2%	€1.036	€2.946	Unknown

W4 retention is standardized over signup months and platforms. This reduces differences in composition without demonstrating a causal channel effect. Fixed-horizon revenue uses the same January-June 2025 signups, including inactive users. It approximates observed-horizon value, not lifetime LTV or margin. Mean paid revenue at 52 weeks does not cover CAC. Other channel costs are unknown, not zero.

Data-cleaning log

Rules are applied in the order below. Affected rows do not sum to distinct people. Raw files are retained. The detailed log remains in docs/cleaning_log.csv.

Issue	Rows	Treatment
Exact duplicate user rows	124	Remove duplicates
Exact duplicate activity rows	1,663	Remove duplicates
Signup dates containing /	311	Read as day/month/year
Push variant labels	211	Standardize variant labels
Missing ages	657	Keep as unknown age
Negative durations	95	Set duration missing, retain row
Durations above 10,080 min	100	Set duration missing, retain row
Durations equal to 10,080 min	105	Flag and exclude suspicious duration
May 2025 paid spend	1	Estimate €1,115 between April and June

After cleaning: 20,785 users, 332,736 activity rows and 18 spend rows. No account is removed without a reliable flag. Missing activity does not prove an uninstall.

Model and scenarios

The model sums weekly cohorts. For each channel, next active users equal existing active users multiplied by a continuation rate, plus new signups. Paid uses budget/CAC. Organic and podcast extend their historical signup trends. Their costs are not invented. Listening and impressions use historical mean consumption, separating new signups from other active users.

Model selection and backtest

On internal validation from 28 April to 29 June, listening error is 1.57% with one continuation rate per channel, versus 2.54% with the tested age bands. We select the simple model, then fix its structure before the final test.

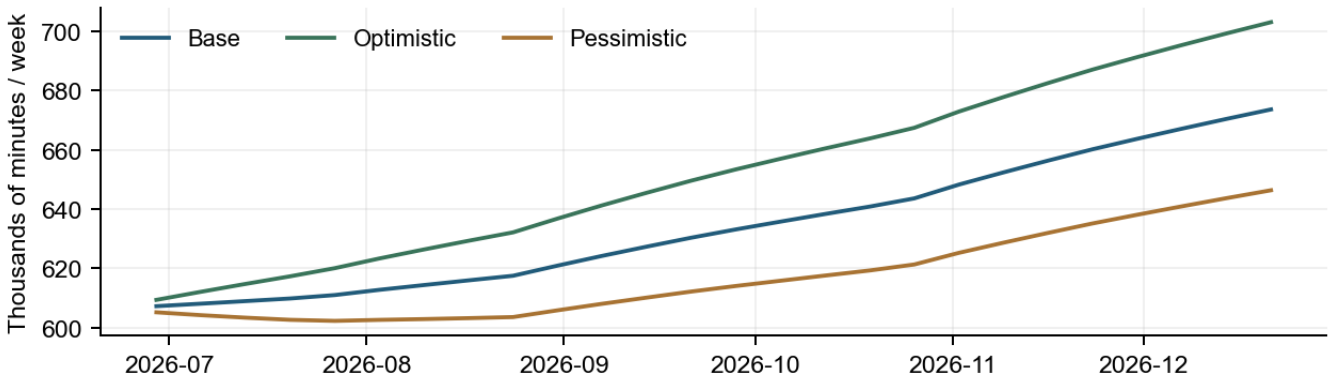
Second half of 2025	Model error	Linear trend error
Active users	3.99%	14.26%
Listening	3.85%	14.27%
Ad revenue	3.87%	14.22%

Error is the sum of absolute differences divided by the actual total across 26 weeks. Parameters stop at 29 June. Realized budgets are known inputs. Test-period signups and activity are not used for fitting. Listening bias is -3.58%. The last week has 4,844 forecast active users versus 5,229 observed.

Conditional forecast over twenty-six weeks

From 29 June to 27 December 2026, scenarios start from 7,468 observed active users and retain parameters validated on 2025. All assume normal notifications and a €1,630 monthly paid budget. Optimistic: CAC and churn probability -5%. Pessimistic: +5%. Other parameters remain identical. No probabilities are assigned to these cases.

Scenario	Cumulative listening million minutes	Final active users	Ad revenue
Base	16.49	8,294	€20,982
Optimistic	16.97	8,658	€21,587
Pessimistic	16.05	7,957	€20,412



Total paid spend is about €9,678 over these 26 weeks. Other costs are unavailable, so revenue minus this budget is not net profit. A paid effect on organic/podcast is possible but unidentified. No additional effect is quantified as measured.

Next experiment and limitations

Proposed A/B test plan

We propose an experiment among new signups who enabled notifications before assignment. Control keeps normal notifications. The variant caps sends at three per week, uses neutral copy and lets users set their frequency. User-level random assignment avoids treating sessions from the same person differently.

Each participant is followed for 84 days from assignment. The primary outcome is cumulative listening minutes per assigned person, including those who become inactive. Daily instrumentation should record listening and measurement errors. Missing values will not automatically become zero. Tracking parameters and missing-data treatment will be fixed before launch.

Sample size planning targets a +5% listening effect without promising it. The retention guardrail allows at most a two-percentage-point deterioration at the end of follow-up. The plan uses 80% power and a 5% significance level. Historical estimates give about 7,800 people per group, or 15,600 overall. The guardrail requires the larger sample. This approximation must be recalculated after an instrumentation pilot.

Recruitment continues until the fixed sample is reached, then the last participant completes twelve weeks. The full experiment therefore lasts longer than twelve weeks. Eligible signup volume and costs may make recruitment lengthy. We do not propose a shorter, underpowered test just to obtain a quick answer.

The proposed decision rule requires a listening gain whose interval excludes zero and a passed retention guardrail. Opt-outs and complaints are also monitored. No opportunistic stopping on interim performance is planned. A technical or consent issue triggers an immediate review. This variant is a new experiment. Its effects are not inferred from the aggressive version.

Decision limitations

The existing experiment covers new signups and twelve complete calendar weeks after their signup week. It does not establish effects on existing users or lifetime outcomes. Notification fatigue is a possible explanation, without direct measures of satisfaction, uninstalls or notification opt-outs.

The chosen North Star is total known listening minutes per week. Active users, median listening and retention accompany it: volume can rise solely because the audience grows. Scenarios assume stable behavior and are not statistical intervals. A six-month backtest does not validate lifetime LTV.

Sources and calculations

[1] Echo Project data pack, IMT 2026, context, columns and CPM.
[2] users.csv, weekly_activity.csv and marketing_spend.csv, extracted on 30 June 2026.
[3] G06_Echo_model.ipynb and docs/cleaning_log.csv, reproducible calculations, comparisons and assumptions.
[4] Growth modeling class and IMT Growth Model Simulator, budget/CAC mechanism and scenarios.